Lecture 18

Heisenberg group and the Weyl relations

Objectives. *The Heisenberg group is the simplest noncommutative nilpotent Lie group. This lecture builds its Schrödinger representation on $L^2(\mathbb{R})$ from translations and modulations, and proves the Weyl relations, which describe how translations and modulations commute up to a phase. On this representation, the bracket $[X, Y] = Z$ becomes the commutation relation $[Q, P] = i\hbar$ of quantum mechanics. We then treat the Stone–von Neumann theorem. It says that a fixed nonzero central character, which records the scalars by which the centre acts, determines one irreducible unitary representation up to unitary equivalence. We prove the irreducibility of the Schrödinger representation from an explicit rank-one operator, and we cite the uniqueness. For central character zero, the irreducibles are one-dimensional characters, and the translation representation on $L^2(\mathbb{R})$ contains none of them.*

By the end of this lecture you will be able to:

- Compute products, inverses, commutators, the exponential map and Ad in the Heisenberg group.
- Prove that the Schrödinger formula is a strongly continuous unitary representation satisfying the Weyl relations, and compute its derivative on $C_c^\infty(\mathbb{R})$.
- State the Stone–von Neumann theorem with its hypotheses, and turn a Weyl pair into a representation to which it applies.
- Compute a Gaussian matrix coefficient, use it to prove that Π_λ is irreducible for $\lambda \neq 0$, and contrast nonzero and zero central character.

18.1 The group law

Definition 18.1. The *Heisenberg group* H is the set of real matrices

$$(a, b, c) := \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}, \qquad a, b, c \in \mathbb{R},$$

with matrix multiplication.

Multiplying the matrices gives

$$(a, b, c)(a', b', c') = (a + a',\ b + b',\ c + c' + ab'). \tag{18.1}$$

The identity is $(0,0,0)$, and

$$(a,b,c)^{-1} = (-a,-b,-c+ab).$$

Indeed, the third coordinate of $(a,b,c)(-a,-b,-c+ab)$ is $c-c+ab-ab=0$. The group H is a closed subgroup of $\mathrm{GL}_3(\mathbb{R})$, hence a Lie group by the closed-subgroup theorem of Lecture 3. The coordinates identify H with $\mathbb{R}^3$, so H is connected and simply connected.

Swapping the factors in the group law (18.1) changes only the last coordinate. For $g=(a,b,c)$ and $g'=(a',b',c')$,

$$gg' = (0,0,ab'-a'b)\,g'g. \tag{18.2}$$

By the group law (18.1), each $(0,0,d)$ commutes with every element. So the commutation formula (18.2) shows that g is central exactly when $ab'-a'b=0$ for all a',b', that is, when $a=b=0$. The centre is therefore

$$Z(H) = \{(0,0,c) : c\in\mathbb{R}\} \cong \mathbb{R},$$

and $(a,b,c)Z(H)\mapsto(a,b)$ identifies $H/Z(H)$ with $\mathbb{R}^2$.

The Lie algebra $\mathfrak{h}$ has basis $X=E_{12}$, $Y=E_{23}$, $Z=E_{13}$, where E_{ij} are matrix units. Then

$$[X,Y]=Z, \qquad [X,Z]=[Y,Z]=0.$$

Let $W=xX+yY+zZ$. Multiplying matrix units gives $W^2=xyZ$ and $W^3=0$. So the exponential series of W stops after the quadratic term:

$$\exp(xX+yY+zZ) = I+W+\tfrac12 W^2 = \big(x,\ y,\ z+\tfrac12 xy\big). \tag{18.3}$$

So $\exp\colon\mathfrak{h}\to H$ is a diffeomorphism with inverse $(a,b,c)\mapsto aX+bY+(c-\frac12 ab)Z$. In particular $\exp(sX)=(s,0,0)$, $\exp(sY)=(0,s,0)$ and $\exp(sZ)=(0,0,s)$.

For the adjoint action we show one representative step. By the group law (18.1),

$$g\exp(sX)g^{-1} = (a+s,b,c)(-a,-b,-c+ab) = (s,0,-sb) = \exp\big(s(X-bZ)\big).$$

Thus $\mathrm{Ad}(g)X = X-bZ$. In the same way $\mathrm{Ad}(g)Y = Y+aZ$ and $\mathrm{Ad}(g)Z=Z$. In the ordered basis (X,Y,Z),

$$D_{\mathrm{Ad}}(g) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -b & a & 1 \end{pmatrix}, \qquad \det D_{\mathrm{Ad}}(g) = 1.$$

With the convention $\Delta(g)=|\det\mathrm{Ad}(g)|^{-1}$ of Lecture 16, H is unimodular. Directly, left and right translations are polynomial maps of $\mathbb{R}^3$ with triangular Jacobian matrices and ones on the diagonal. So Lebesgue measure $da\,db\,dc$ is a left and right Haar measure.

18.2 The Schrödinger representation

We use $L^2(\mathbb{R})$ with $\langle f,h\rangle = \int_{\mathbb{R}} \overline{f(t)}\,h(t)\,dt$. We use the integration rules stated at the end of Lecture 16, in its section on locally compact groups: the identification of $L^2(\mathbb{R})$ with classes of functions, Tonelli and Fubini, dominated convergence and change of variables. To differentiate under an integral, we apply dominated convergence to the difference

quotients, using a common integrable bound for the derivative. We also use two standard facts. First, Lebesgue integrals are invariant under $t \mapsto t + a$. Second, the space $C_c(\mathbb{R})$ of continuous compactly supported functions, and its subspace $C_c^\infty(\mathbb{R})$ of smooth ones, are dense in $L^2(\mathbb{R})$.

Definition 18.2. Fix $\lambda \in \mathbb{R}$. For $g = (a, b, c) \in H$ and $f \in L^2(\mathbb{R})$, the *Schrödinger operator* $\Pi_\lambda(g)$ is defined by

$$(\Pi_\lambda(a, b, c)f)(t) = e^{i\lambda(c+bt)} f(t + a). \tag{18.4}$$

The *translations* $U(a)$ and *modulations* $V_\lambda(b)$ are defined by

$$(U(a)f)(t) = f(t + a), \qquad (V_\lambda(b)f)(t) = e^{i\lambda bt} f(t).$$

By the definitions, the Schrödinger operators factor as

$$\Pi_\lambda(a, b, c) = e^{i\lambda c}\, V_\lambda(b)\, U(a).$$

This matches the factorisation of group elements

$$(a, b, c) = (0, 0, c)(0, b, 0)(a, 0, 0).$$

Our $U(a)$ is the translation by $-a$ in the convention of Lecture 15, and $V_1(b)$ is the modulation $M(b)$ of Exercise 15.5. So at $\lambda = 1$ the Weyl relations below are the relation of Exercise 15.5(d).

Proposition 18.3. *For every $\lambda \in \mathbb{R}$, Π_λ is a strongly continuous unitary representation of H on $L^2(\mathbb{R})$, and $\Pi_\lambda(0, 0, c) = e^{i\lambda c} I$. The* Weyl relations *hold:*

$$U(a)V_\lambda(b) = e^{i\lambda ab} V_\lambda(b)U(a), \qquad a, b \in \mathbb{R}. \tag{18.5}$$

Proof. Homomorphism. Let $g = (a, b, c)$ and $g' = (a', b', c')$. Applying the defining formula (18.4) twice,

$$\begin{aligned}(\Pi_\lambda(g)\Pi_\lambda(g')f)(t) &= e^{i\lambda(c+bt)}(\Pi_\lambda(g')f)(t + a)\\ &= e^{i\lambda(c+bt)} e^{i\lambda(c'+b'(t+a))} f(t + a + a')\\ &= e^{i\lambda(c+c'+ab'+(b+b')t)} f(t + a + a').\end{aligned}$$

By the group law (18.1), this is $(\Pi_\lambda(gg')f)(t)$. Also $\Pi_\lambda(0, 0, 0) = I$.

Unitarity. Since $|e^{i\lambda(c+bt)}| = 1$, translation invariance gives $\|\Pi_\lambda(g)f\|^2 = \int |f(t+a)|^2 dt = \|f\|^2$. Each $\Pi_\lambda(g)$ is an isometry with inverse $\Pi_\lambda(g^{-1})$, hence unitary.

Centre and Weyl relations. The first claim is the defining formula (18.4) with $a = b = 0$. For the second,

$$(U(a)V_\lambda(b)f)(t) = (V_\lambda(b)f)(t + a) = e^{i\lambda b(t+a)} f(t + a) = e^{i\lambda ab}(V_\lambda(b)U(a)f)(t).$$

Strong continuity. This is the only analytic step. First let $f \in C_c(\mathbb{R})$ vanish outside $[-R, R]$, and let $|a| \le 1$. Then $\Pi_\lambda(g)f - f$ vanishes outside $[-R-1, R+1]$. On that interval, $|e^{i\theta} - 1| \le |\theta|$ gives

$$|(\Pi_\lambda(g)f)(t) - f(t)| \le |\lambda|\big(|c| + (R + 1)|b|\big)\|f\|_\infty + \omega_f(|a|),$$

where $\omega_f(\delta) = \sup\{|f(t+s)-f(t)| : t \in \mathbb{R},\ |s| \le \delta\}$. Uniform continuity of f gives $\omega_f(\delta) \to 0$ as $\delta \to 0$. Integrating the square over an interval of length $2R+2$,

$$\|\Pi_\lambda(g)f - f\| \le (2R+2)^{1/2}\Big(|\lambda|\big(|c| + (R+1)|b|\big)\|f\|_\infty + \omega_f(|a|)\Big) \longrightarrow 0 \quad (g \to e).$$

Now let $f \in L^2(\mathbb{R})$ and $\varepsilon > 0$. Choose $h \in C_c(\mathbb{R})$ with $\|f-h\| < \varepsilon$. Since $\Pi_\lambda(g)$ is unitary,

$$\|\Pi_\lambda(g)f - f\| \le \|\Pi_\lambda(g)(f-h)\| + \|\Pi_\lambda(g)h - h\| + \|h - f\| < 2\varepsilon + \|\Pi_\lambda(g)h - h\|.$$

So $\limsup_{g\to e} \|\Pi_\lambda(g)f - f\| \le 2\varepsilon$ for every ε. Finally, $\|\Pi_\lambda(g)f - \Pi_\lambda(g_0)f\| = \|\Pi_\lambda(g_0^{-1}g)f - f\|$, and $g_0^{-1}g \to e$ as $g \to g_0$. ∎

Combining the commutation formula (18.2) with the scalar action of the centre in Proposition 18.3 gives

$$\Pi_\lambda(g)\Pi_\lambda(g') = e^{i\lambda(ab' - a'b)}\,\Pi_\lambda(g')\Pi_\lambda(g).$$

For $\lambda \neq 0$ the operators need not commute. For $\lambda = 0$ they commute.

This answers the question left open by the Heisenberg multiplier example of Lecture 13. There $\omega(x,y) = x_1y_2 - x_2y_1$ on $\mathbb{R}^2$ and $c(x,y) = e^{i\omega(x,y)/2}$. For $x \in \mathbb{R}^2$ put $U_x = \Pi_1(\exp(x_1X + x_2Y))$. The exponential formula (18.3) and the group law give

$$U_xU_y = e^{i\omega(x,y)/2}\,U_{x+y}.$$

So the multiplier c is realised on $L^2(\mathbb{R})$. By Exercise 18.4(d), Π_1 has no nonzero finite-dimensional invariant subspace, as that example requires. The extension $E = \mathbb{R}^2 \times \mathbb{T}$ of Exercise 13.5 is the quotient H/Γ with $\Gamma = \{(0,0,2\pi k) : k \in \mathbb{Z}\}$, through the map $(a,b,c) \mapsto ((a,b), e^{i(c-ab/2)})$.

The exponent $ab' - a'b$ is a signed area in the plane $H/Z(H)$, as Figure 18.1 shows. Multiply on the right by $(a,0,0)$ and then by $(0,b,0)$. This route ends at (a,b,ab). The other route ends at $(a,b,0)$. The central coordinates differ by the area ab of the rectangle, and this is the phase in the Weyl relations. Similarly, the four steps g, g', g^{-1}, g'^{-1} form a closed loop in the plane, and by (18.2) they end at $(0,0,ab'-a'b)$. The loop runs counterclockwise when $ab' - a'b > 0$. The operator products in the figure name these group products. As operators they act on a vector from the right, so the order of action is the reverse of the order of the route.

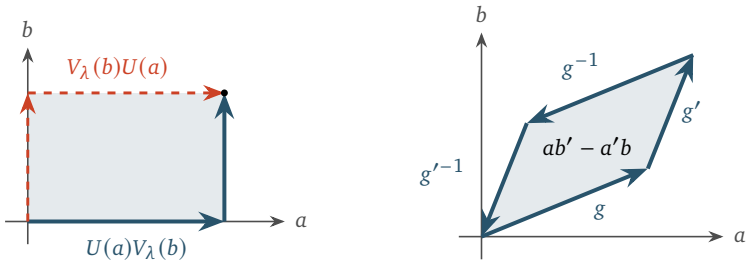


Figure 18.1: Routes in the plane $H/Z(H)$, with each step a multiplication on the right. In the first panel, two routes join e to the same point of the plane, and the rectangle lies between them. In the second panel, the loop of a commutator encloses the parallelogram on (a,b) and (a',b'), and its signed area is $ab' - a'b$.

Remark 18.4. Let $A = \{(0, b, c)\}$, a closed abelian normal subgroup, with character $\chi_\lambda(0, b, c) = e^{i\lambda c}$. Exercise 18.5 shows that the representation unitarily induced from χ_λ, as in Lecture 17, is equivalent to Π_λ.

Next we differentiate. Write $\mathcal{D} = C_c^\infty(\mathbb{R})$ and tf for the function $t \mapsto tf(t)$.

Proposition 18.5. *Each $\Pi_\lambda(g)$ maps $\mathcal{D}$ into $\mathcal{D}$. For $f \in \mathcal{D}$ and $W \in \{X, Y, Z\}$, the limit*

$$d\Pi_\lambda(W)f = \lim_{s \to 0} s^{-1}(\Pi_\lambda(\exp sW)f - f)$$

exists in $L^2(\mathbb{R})$, and

$$d\Pi_\lambda(X)f = f', \qquad d\Pi_\lambda(Y)f = i\lambda t f, \qquad d\Pi_\lambda(Z)f = i\lambda f.$$

On $\mathcal{D}$ we have $[d\Pi_\lambda(X), d\Pi_\lambda(Y)] = d\Pi_\lambda(Z) = d\Pi_\lambda([X, Y])$.

Proof. Translation and multiplication by the smooth function $e^{i\lambda(c+bt)}$ preserve smoothness and compact support. Let f vanish outside $[-R, R]$ and let $0 < |s| \le 1$. For X, Taylor's theorem gives

$$\left| \frac{f(t+s) - f(t)}{s} - f'(t) \right| \le \tfrac{1}{2}|s| \, \|f''\|_\infty,$$

and the left side vanishes outside $[-R-1, R+1]$. Its L^2 norm is at most $(2R+2)^{1/2}\tfrac{1}{2}|s|\|f''\|_\infty \to 0$. For Y, the bound $|e^{i\theta} - 1 - i\theta| \le \theta^2/2$ gives

$$\left| \frac{e^{i\lambda st} - 1}{s} - i\lambda t \right| |f(t)| \le \tfrac{1}{2}\lambda^2 |s| R^2 |f(t)|,$$

whose L^2 norm tends to 0. For Z, $(e^{i\lambda s} - 1)/s \to i\lambda$. Finally,

$$[d\Pi_\lambda(X), d\Pi_\lambda(Y)]f = (i\lambda t f)' - i\lambda t f' = i\lambda f = d\Pi_\lambda(Z)f. \qquad \blacksquare$$

Remark 18.6. Since Π_λ is strongly continuous (Proposition 18.3), $s \mapsto \Pi_\lambda(\exp sW)$ is a strongly continuous one-parameter unitary group. Its generator B_W, as defined in Lecture 15, has domain exactly the vectors f at which $s \mapsto \Pi_\lambda(\exp sW)f$ is differentiable at $s = 0$ in $L^2(\mathbb{R})$. Stone's theorem (Lecture 15) shows that B_W is skew-adjoint. So the physical generator $A_W = iB_W$ is self-adjoint and satisfies $\Pi_\lambda(\exp sW) = \exp(-isA_W)$. The derivative formulas of Proposition 18.5 show that $\mathcal{D}$ lies in the domain of A_W and that on $\mathcal{D}$

$$A_X f = if', \qquad A_Y f = -\lambda t f, \qquad A_Z f = -\lambda f.$$

Since $\mathcal{D}$ is dense and invariant under each $\Pi_\lambda(\exp sW)$, the dense invariant-domain criterion of Stone's theorem in Lecture 15 shows that B_W is the closure of $d\Pi_\lambda(W)$ on $\mathcal{D}$.

In quantum mechanics a particle on a line has state space $L^2(\mathbb{R})$. Take $\lambda = 1/\hbar$, where $\hbar > 0$ is the reduced Planck constant. Put $P = -\hbar A_X$ (momentum) and $Q = -\hbar A_Y$ (position). The minus sign in P is there because $U(a)$ is the translation by $-a$. On $\mathcal{D}$, $Qf = tf$, $Pf = -i\hbar f'$ and $[Q, P] = i\hbar$.

In terms of these operators, $U(a) = \exp(iaP/\hbar)$ and $V_\lambda(b) = \exp(ibQ/\hbar)$. So the Weyl relations (18.5) are the bounded, exponentiated form of the commutation relation.

We never use the commutation relation as a hypothesis. A commutation relation of unbounded operators on a dense domain does not by itself imply the Weyl relations. For example, on $L^2(0,1)$ take $Qf = tf$ and $P = -i\hbar\, d/dt$ with periodic boundary conditions. Both operators are self-adjoint, and $[Q,P] = i\hbar$ on $C_c^\infty(0,1)$. Here $\exp(iP/\hbar) = I$, since P has the eigenvectors $e^{2\pi i n t}$ with eigenvalues $2\pi\hbar n$. The Weyl relation at $a = 1$ would then give $I = e^{ib/\hbar} I$ for every $b \in \mathbb{R}$, which is false (Hall, *Quantum Theory for Mathematicians*, Section 14.1). Exercise 18.3 shows that no bounded operators on a nonzero Hilbert space satisfy $[Q,P] = i\hbar I$.

18.3 The Stone–von Neumann theorem

Definition 18.7. A unitary representation ρ of H on a Hilbert space $\mathcal{H}$ has *central character* $\lambda \in \mathbb{R}$ if $\rho(0,0,c) = e^{i\lambda c} I$ for all $c \in \mathbb{R}$.

By the centre statement in Proposition 18.3, the representation Π_λ has central character λ. Now let ρ be an irreducible unitary representation of H. Schur's lemma for unitary representations (Kirillov, *Lectures on the Orbit Method*, 2004, Appendix V) makes each central operator $\rho(0,0,c)$ a scalar. These scalars form a continuous homomorphism $\mathbb{R} \to \mathrm{U}(1)$, hence have the form $e^{i\lambda c}$. So every irreducible unitary representation has a central character. Reducible ones need not, for example $\Pi_1 \oplus \Pi_2$.

Lemma 18.8. *Let σ and τ be strongly continuous one-parameter unitary groups on a Hilbert space $\mathcal{H}$, and let $\lambda \in \mathbb{R}$. Suppose $\sigma(a)\tau(b) = e^{i\lambda ab}\tau(b)\sigma(a)$ for all $a, b \in \mathbb{R}$. Then $\rho(a,b,c) = e^{i\lambda c}\tau(b)\sigma(a)$ defines a strongly continuous unitary representation ρ of H on $\mathcal{H}$ with central character λ.*

Proof. Move $\sigma(a)$ past $\tau(b')$ using the hypothesis:

$$\rho(g)\rho(g') = e^{i\lambda(c+c')}\tau(b)\sigma(a)\tau(b')\sigma(a') = e^{i\lambda(c+c'+ab')}\tau(b+b')\sigma(a+a') = \rho(gg').$$

Each $\rho(g)$ is unitary, and $\rho(0,0,c) = e^{i\lambda c} I$. For continuity at (a_0,b_0,c_0), let $f \in \mathcal{H}$ and put $h = \sigma(a_0)f$. Then

$$\|\tau(b)\sigma(a)f - \tau(b_0)\sigma(a_0)f\| \le \|\sigma(a)f - \sigma(a_0)f\| + \|\tau(b)h - \tau(b_0)h\| \to 0,$$

and the scalar factor $e^{i\lambda c}$ is continuous. ∎

We call a pair (σ,τ) satisfying the hypotheses of Lemma 18.8 a *Weyl pair* with parameter λ. By the Weyl relations (18.5) and the strong continuity in Proposition 18.3, the translations and modulations form a Weyl pair (U, V_λ) with parameter λ. Two Weyl pairs are *equivalent* when a single unitary operator from the first space to the second conjugates each group of the first pair into the corresponding group of the second.

Theorem 18.9 (Stone–von Neumann). *Fix $\lambda \neq 0$.*

(a) The Schrödinger representation Π_λ is irreducible.

(b) Let ρ be a strongly continuous unitary representation of H on a nonzero separable Hilbert space $\mathcal{H}$ with central character λ. Then $\mathcal{H}$ is a Hilbert direct sum of mutually orthogonal closed ρ-invariant subspaces. On each summand ρ is unitarily equivalent to Π_λ.

We prove part (a) in § 18.4, after a kernel computation that does not use this theorem. We cite part (b), the uniqueness statement, from Kirillov, *Lectures on the Orbit Method*, Chapter 2. The proof of (a) does not prove (b).

The Hilbert space of an irreducible representation of H is automatically separable. By irreducibility, every nonzero vector is cyclic. By strong continuity, the closed span of the orbit of such a vector is already generated by its orbit under a countable dense subset of H. This closed span is the whole space, and it is separable.

The uniqueness statement in the Stone–von Neumann theorem, Theorem 18.9, has the following consequences.

- Every irreducible unitary representation of H with central character $\lambda \neq 0$ is unitarily equivalent to Π_λ.
- By the lemma on Weyl pairs, Lemma 18.8, every Weyl pair with parameter $\lambda \neq 0$ on a nonzero separable space is equivalent to a direct sum of copies of (U, V_λ). In particular, every Weyl pair with parameter $1/\hbar$ on a nonzero separable space is unitarily equivalent to a direct sum of copies of $(\exp(iaP/\hbar), \exp(ibQ/\hbar))$.
- Distinct values of λ give inequivalent representations Π_λ (Exercise 18.6(c) on inequivalence).

In quantum mechanics, the first two consequences say that every irreducible Weyl pair with parameter $1/\hbar$ is the Schrödinger pair, which is the precise form of the statement that matrix mechanics and wave mechanics agree.

A standard proof of (b), going back to von Neumann, turns on one step. Let ρ be a representation as in the uniqueness statement (b) of Theorem 18.9. For $f, h \in \mathcal{H}$, consider the sesquilinear form

$$B(f,h) = \frac{|\lambda|}{2\pi} \int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \, \big\langle f, \rho(\exp(aX + bY))h \big\rangle \, da\, db.$$

The Gaussian weight has integral $4\pi/|\lambda|$, so $|B(f,h)| \leq 2\|f\|\|h\|$. Hence the Riesz representation theorem for bounded sesquilinear forms gives a bounded operator E on $\mathcal{H}$ with $\|E\| \leq 2$ and $B(f,h) = \langle f, Eh\rangle$ for all $f, h \in \mathcal{H}$.

That proof shows, from the Weyl relations and strong continuity, that E is a nonzero orthogonal projection. Each unit vector $v \in E\mathcal{H}$ then generates one copy of Π_λ. In that copy, v corresponds to the Gaussian of the next section, § 18.4. The weight is that Gaussian's matrix coefficient, given by (18.6) at $g = \exp(aX + bY)$.

18.4 Gaussian matrix coefficients and central character zero

Let $\varphi(t) = \pi^{-1/4} e^{-t^2/2}$. Since $\int e^{-t^2} dt = \sqrt{\pi}$, $\|\varphi\| = 1$. We compute $\langle \varphi, \Pi_\lambda(g)\varphi\rangle$ for $g = (a, b, c)$. We use the calculus fact

$$\int_{\mathbb{R}} e^{-(t+w)^2} dt = \sqrt{\pi} \qquad (w \in \mathbb{C}).$$

For real w this is a translation. Write $w = u + iv$. The v-derivative of the integrand is $i\,\frac{d}{dt} e^{-(t+w)^2}$, which integrates to 0. Differentiation under the integral is justified on $|v| \leq V$

by the common integrable bound $2(|t+u|+V)e^{V^2}e^{-(t+u)^2}$ for the derivative. So the integral does not depend on v.

By the defining formula (18.4), and since φ is real,

$$\langle \varphi, \Pi_\lambda(g)\varphi\rangle = \pi^{-1/2} e^{i\lambda c} \int_{\mathbb{R}} e^{-t^2/2-(t+a)^2/2+i\lambda b t}\, dt.$$

Complete the square with $w = \frac{1}{2}(a - i\lambda b)$:

$$-\tfrac{t^2}{2} - \tfrac{(t+a)^2}{2} + i\lambda b t = -t^2 - (a - i\lambda b)t - \tfrac{a^2}{2} = -(t+w)^2 + w^2 - \tfrac{a^2}{2},$$

$$w^2 - \tfrac{a^2}{2} = \tfrac{1}{4}\big(a^2 - 2i\lambda ab - \lambda^2 b^2\big) - \tfrac{a^2}{2} = -\tfrac{1}{4}\big(a^2 + \lambda^2 b^2\big) - \tfrac{i\lambda ab}{2}.$$

The integral is $\sqrt{\pi}$ times $e^{w^2 - a^2/2}$. Therefore

$$\langle \varphi, \Pi_\lambda(a,b,c)\varphi\rangle = e^{i\lambda(c - ab/2)}\, e^{-(a^2+\lambda^2 b^2)/4}. \tag{18.6}$$

We make two checks. First, at $g = e$ the value is $1 = \|\varphi\|^2$. Second, replacing g by $g^{-1} = (-a, -b, -c + ab)$ conjugates the value, as unitarity requires.

By the exponential formula (18.3), $c - ab/2$ is the Z-coordinate of $\log g$, where log is the inverse of the exponential map. Hence at $\exp(xX + yY + zZ)$ the coefficient is $e^{i\lambda z} e^{-(x^2+\lambda^2 y^2)/4}$.

The modulus depends only on the image (a, b) in $H/Z(H) \cong \mathbb{R}^2$. For $\lambda \neq 0$ it decays in every direction, and

$$\int_{\mathbb{R}^2} \big|\langle \varphi, \Pi_\lambda(a,b,c)\varphi\rangle\big|^2 da\, db = \int e^{-a^2/2} da \int e^{-\lambda^2 b^2/2} db = \sqrt{2\pi} \cdot \frac{\sqrt{2\pi}}{|\lambda|} = \frac{2\pi}{|\lambda|}.$$

Its reciprocal $|\lambda|/(2\pi)$ is the normalising constant in the form B above.

Compare Schur orthogonality (Lecture 5). For a unit vector v in an irreducible unitary representation of dimension d of a compact group, $\int |\langle v, \Pi(g)v\rangle|^2\, dg = 1/d$, and $d \int \overline{\langle v, \Pi(g)v\rangle}\, \Pi(g)\, dg$ is the orthogonal projection onto $\mathbb{C}v$. The form B has the same shape on $H/Z(H)$, with φ in place of v. The number $|\lambda|/(2\pi)$ takes the place of d. Only the shape can be compared, because the measure here is $da\, db$ and not a normalised integral.

The operator E defined after the Stone–von Neumann theorem can be computed explicitly in this model, for $\lambda \neq 0$. The computation does not use that theorem. Write E_λ for that operator with $\rho = \Pi_\lambda$. Since

$$\Pi_\lambda(\exp(aX + bY))f(t) = e^{i\lambda(bt + ab/2)} f(t + a),$$

we integrate first over b. For $f, h \in C_c(\mathbb{R})$, Tonelli, Cauchy–Schwarz in t and translation invariance give, as at the end of Lecture 16,

$$\int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \int_{\mathbb{R}} |h(t)|\, |f(t+a)|\, dt\, da\, db \le \frac{4\pi}{|\lambda|} \|f\| \|h\|.$$

Thus Fubini's theorem identifies the following kernel calculation with the form B defining E_λ. The Gaussian Fourier integral, obtained by the same square completion as above, gives

$$\int_{\mathbb{R}} e^{-\lambda^2 b^2/4} e^{i\lambda b(t + a/2)}\, db = \frac{2\sqrt{\pi}}{|\lambda|} e^{-(t+a/2)^2}.$$

Consequently, for $f \in C_c(\mathbb{R})$,

$$(E_\lambda f)(t) = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-a^2/4-(t+a/2)^2} f(t+a)\, da.$$

Set $u = t + a$. The exponent simplifies to

$$-\frac{(u-t)^2}{4} - \frac{(t+u)^2}{4} = -\frac{t^2+u^2}{2}.$$

We therefore obtain

$$(E_\lambda f)(t) = \varphi(t) \int_{\mathbb{R}} \overline{\varphi(u)} f(u)\, du = \varphi(t)\langle \varphi, f\rangle.$$

Both operators are bounded, so density extends the identity to all $L^2(\mathbb{R})$. Thus E_λ is exactly the orthogonal projection onto the Gaussian line. For a general representation ρ with central character $\lambda \neq 0$, proving that the operator E is a nonzero projection is the key step of the standard proof of the uniqueness statement. Here its rank-one integral kernel is visible. In quantum mechanics, with $\lambda = \hbar = 1$, the function φ is the ground state of the oscillator $\frac{1}{2}(P^2 + Q^2)$, the vector $\Pi_1(g)\varphi$ is a coherent state, and E_1 is the projection on the ground state.

We can now prove part (a) of the theorem. The rank of E_λ forces every closed invariant subspace to contain φ or to be orthogonal to it, and the orbit of φ spans a dense subspace.

Proof of Theorem 18.9(a). Fix $\lambda \neq 0$, and let $W \subset L^2(\mathbb{R})$ be a closed invariant subspace.

The orbit of φ spans a dense subspace. Let $f \in L^2(\mathbb{R})$ be orthogonal to $\Pi_\lambda(g)\varphi$ for all $g \in H$. Taking $g = (0, b, 0)$ gives

$$\int_{\mathbb{R}} \overline{f(t)}\, \varphi(t)\, e^{i\lambda b t}\, dt = 0 \qquad \text{for every } b \in \mathbb{R}.$$

By the Cauchy–Schwarz inequality, $u = \bar{f}\varphi$ is integrable. We cite the injectivity of the Fourier transform on $L^1(\mathbb{R})$: if u is integrable and $\int_{\mathbb{R}} u(t)e^{ist}\, dt = 0$ for every $s \in \mathbb{R}$, then $u = 0$ almost everywhere (Folland, *Real Analysis*, second edition, Chapter 8). Since $\lambda \neq 0$, the number $s = \lambda b$ runs over $\mathbb{R}$. So $\bar{f}\varphi = 0$ almost everywhere. Since $\varphi > 0$ everywhere, $f = 0$.

The operator E_λ commutes with the projection onto W. Since $\Pi_\lambda(g)^* = \Pi_\lambda(g^{-1})$, the subspace $W^\perp$ is also invariant. Let $f \in W$ and $h \in W^\perp$. Then $\langle h, \Pi_\lambda(g)f\rangle = 0$ for all $g \in H$. The form defining E_λ therefore gives $\langle h, E_\lambda f\rangle = 0$. Hence $E_\lambda W \subset W$. The same argument with W and $W^\perp$ exchanged gives $E_\lambda W^\perp \subset W^\perp$. So E_λ commutes with the orthogonal projection P_W onto W.

Conclusion. Since $E_\lambda \varphi = \varphi$, we get $E_\lambda P_W \varphi = P_W E_\lambda \varphi = P_W \varphi$. So $P_W\varphi$ lies in the range $\mathbb{C}\varphi$ of E_λ, and $P_W\varphi = \kappa\varphi$ for some $\kappa \in \mathbb{C}$. Since $P_W^2 = P_W$, we have $\kappa = 0$ or $\kappa = 1$. If $\kappa = 1$, then $\varphi \in W$, and the first step gives $W = L^2(\mathbb{R})$. If $\kappa = 0$, then $\varphi \in W^\perp$, and the first step gives $W^\perp = L^2(\mathbb{R})$, so $W = 0$. ∎

Now set $\lambda = 0$. The defining formula (18.4) becomes pure translation, $(\Pi_0(a, b, c)f)(t) = f(t+a)$. The modulations become trivial, and the Gaussian coefficient (18.6) becomes $e^{-a^2/4}$. This does not decay in b, and the integral above diverges.

The irreducible unitary representations of H with central character 0 factor through $H/Z(H) \cong \mathbb{R}^2$. By Schur's lemma (Kirillov, Appendix V), irreducible unitary representations of an abelian group are one-dimensional. Hence the irreducible unitary representations of H with central character 0 are the characters

$$\chi_{\alpha,\beta}(a,b,c) = e^{i(\alpha a+\beta b)}, \qquad \alpha,\beta \in \mathbb{R}.$$

Kirillov's Chapter 2 records this together with the Stone–von Neumann theorem, Theorem 18.9, as the full list of irreducibles. Lecture 19 explains the list through coadjoint orbits.

Proposition 18.10. *Π_0 has no one-dimensional invariant subspace. Hence it has no irreducible closed invariant subspace.*

Proof. Suppose $f \neq 0$ spans an invariant line. Then $U(a)f = \mu(a)f$ with $|\mu(a)| = 1$. For each integer n,

$$\int_n^{n+1} |f(t)|^2 dt = \int_0^1 |f(t+n)|^2 dt = \int_0^1 |\mu(n)f(t)|^2 dt = \int_0^1 |f(t)|^2 dt.$$

Summing over $n \in \mathbb{Z}$ gives $\|f\|^2 = \sum_n \int_0^1 |f|^2$. This is finite only if $\int_0^1 |f|^2 = 0$, and then $\|f\| = 0$, a contradiction. An irreducible closed invariant subspace would carry an irreducible representation with central character 0. By the classification above it would be a line. ∎

The pieces one expects in Π_0 are the plane waves $e^{i\alpha t}$, on which $U(a)$ acts by $e^{i\alpha a}$. They are not square integrable. The Fourier transform of Lecture 15 turns $U(a)$ into multiplication by $e^{i\alpha a}$ on $L^2(\mathbb{R}, d\alpha)$. So Π_0 is an integral of the characters $\chi_{\alpha,0}$ rather than a sum. This is called a *direct integral*, and we do not develop it.

The following table compares central character $\lambda \neq 0$ with central character 0.

	central character $\lambda \neq 0$	central character 0
irreducibles	one class, Π_λ	$\chi_{\alpha,\beta}$, $\alpha,\beta \in \mathbb{R}$
dimension	infinite	one
the $L^2(\mathbb{R})$ model	Π_λ itself, irreducible	Π_0 has no irreducible piece
Gaussian coefficient	decays on $H/Z(H)$	constant in b

Self-checks

1. Compute gg' and $g'g$ for $g = (1,2,3)$ and $g' = (4,5,6)$, and check the commutation formula (18.2).
   Answer. $gg' = (5, 7, 3+6+5) = (5,7,14)$ and $g'g = (5,7,9+8) = (5,7,17)$. Here $ab' - a'b = 5 - 8 = -3$, and $(0,0,-3)(5,7,17) = (5,7,14)$.
2. Why are Π_2 and Π_3 inequivalent?
   Answer. An intertwiner T satisfies $Te^{2ic} = e^{3ic}T$ for all c. At $c = \pi$ this reads $T = -T$, so $T = 0$.
3. Compute $\langle \varphi, \Pi_1(2,2,2)\varphi \rangle$ and $\langle \varphi, \Pi_0(0,b,0)\varphi \rangle$.
   Answer. By the Gaussian coefficient (18.6), $c - ab/2 = 0$ and $(a^2+b^2)/4 = 2$, so the first is e^{-2}. The second is 1 for every b.

Summary

The Heisenberg group has centre $Z(H) \cong \mathbb{R}$, and every commutator lies in it. Translations and modulations on $L^2(\mathbb{R})$ form the Schrödinger representation Π_λ, which satisfies the Weyl relations. For $\lambda \neq 0$ the Gaussian form computes an explicit rank-one projection, and this proves that Π_λ is irreducible. The uniqueness statement of the Stone–von Neumann theorem, cited, says that for $\lambda \neq 0$ every strongly continuous unitary representation on a nonzero separable space with central character λ is a Hilbert direct sum of copies of Π_λ, up to unitary equivalence. At $\lambda = 0$ the irreducibles are the characters $\chi_{\alpha,\beta}$, and the translation representation Π_0 contains none of them.

Reading

- Kirillov, *Lectures on the Orbit Method* (2004), Chapter 2: the Heisenberg group, its irreducible unitary representations and the Stone–von Neumann theorem.
- Kirillov, *Lectures on the Orbit Method*, Appendix V: background on unitary representations, including Schur's lemma and induced representations.
- Hall, *Lie Groups, Lie Algebras, and Representations*, second edition (2015), Chapter 1: the Heisenberg group among the examples of matrix Lie groups.